

TORUS ADVANCED (v2.0) | TECHNICAL SPECIFICATION

Concept Developer: Oleksandr Prydatko (Kyiv, Ukraine | saelk@ukr.net) **System Scope:** Emergency & Rescue Wearable AI Wristband featuring Photonic Computing Systems, Acoustic Intelligence, and Stamped Titanium Enclosure

1. GENERAL DESCRIPTION AND ARCHITECTURE

TORUS Advanced is a fully autonomous wearable safety device engineered as a monolithic, strapless four-segment cuff bracelet made of high-strength stamped titanium alloy (Ti-6Al-4V / Grade 5) with a spring-loaded sliding chassis-antenna made of 316L stainless steel. The device is designed for deployment in extreme environments, wilderness expeditions, industrial disaster zones, and military campaigns operating under complete cellular and internet blackout conditions.

The device features a photonic neuromorphic processor, a multispectral biosensor array, an acoustic intelligence suite (sub- and ultrasound), a mini retake hybrid power system, and LoRa/LiFi mesh communication capabilities.

2. PHOTONIC & AI COMPUTING CORE

- **Silicon Photonics Waveguides (PhIC):**
 - Integrated silicon nitride (Si_3N_4) waveguides enabling intra-processor data transmission at light speed.
 - Inter-segment data transfer is performed via flexible printed circuits (FPC) routing both photonic and electrical buses, configured in a serpentine layout (Stretchable Electronics) within the internal cavities of the 316L steel guide rails.
 - Reduces internal thermal dissipation and core-to-core interconnect power consumption by 40–50%.
- **Photonic AI Edge Accelerator:**
 - **Architecture:** Hybrid architecture combining Fabry-Pérot / Photonic Processing Arrays (FPPA) with an ARM Cortex-M55 / Ethos-U55 microcontroller core.
 - **Latency:** Query response and parallel execution of all neural networks in $< 50\text{ ms}$ (compared to 300 ms in the base version).
 - **AI Model Capabilities:**
 - *Proactive Monitoring:* Fall detection, traumatic shock, and hypothermia prediction 3–10 seconds prior to onset.
 - *Predictive Diagnostics:* Biomarker analysis to forecast hypoglycemic attacks, arrhythmia, and cardiac anomalies up to 2 hours in advance.
 - *Interactive AI Rescuer:* On-device survival algorithm retrieval from a local knowledge base (3–4 execution steps) dynamically adapted to real-time threat contexts.

3. SENSOR ARRAY, MICROANALYZER, AND ACOUSTIC INTELLIGENCE

- **Non-Invasive Biochemical Spectroscopy (Internal Circuit):**
 - Multi-wavelength PPG using laser diodes across various optical bands.
 - **Metrics:** Continuous monitoring of blood glucose, lactate level (fatigue/hypoxia), arterial stiffness, heart rate, SpO_2 , ECG, and Galvanic Skin Response (GSR — stress level assessment).
 - **Skin Temperature Sensor:** High-precision measurement ($\pm 0.1^\circ\text{C}$) to register hypothermia or heat stroke.
- **Air Quality Microanalyzer (External Circuit):**
 - Flow-through microchannel integrated into the stamped side segment, sealed with a press-fit IP68/IP69K hydrophobic membrane.
 - Solid-state optoelectronic sensor for TVOC (volatile organic compounds/smoke), CO_2 (rubble/caves), and CO (carbon monoxide).
 - **Operational Logic:** Pulsed polling protocol (once every 2 minutes). When safety thresholds are breached, the AI forces display activation, triggers haptic feedback, and issues an audio alert.
- **Acoustic Intelligence Module (Wideband Acoustic Circuit):**
 - **MEMS Array:** Sealed microphone array covering 1 Hz to 80 kHz (Always-On operation) embedded into structural recesses of the stamped housing.
 - **Infrasound (1–20 Hz):** Early detection of micro-seismic activity, avalanches, landslides, tsunamis, and cave-ins well before visual manifestation.
 - **Ultrasound (20–80 kHz):** Detection of high-pressure gas leaks, structural micro-cracking, and wild animal echolocation/vocalizations.
 - **Acoustic Spectral AI:** Pattern detection for dry combustion (early-stage forest fires), ice cracking, drone/vehicle acoustic signatures, and predator footsteps (bears, wolves) in darkness.
- **Kinematics and Spatial Orientation:**
 - 6-axis IMU (accelerometer + gyroscope) for impact logging, micro-tremor, and fall detection ($>4\text{g}$).
 - 3-axis magnetometer (digital compass) for GPS-denied navigation.

4. HYBRID POWER & MINI RETAKE SYSTEM

- **Sodium-Ion Storage (Na-Ion):**
 - Total capacity $\sim 1450\text{ mAh}$ (energy density of 4.05 Wh , nominal voltage 2.8 V).
 - High resistance to deep discharge (down to 0 V) without degradation; operational capability in extreme sub-zero temperatures (down to -40°C).
 - Cells are enclosed within the stamped side segments (25 mm length).
- **Renewable Energy Harvesting:**
 - Flexible photovoltaic solar cells laminated onto the surface layer of the side segments.
 - **Laser Power Reception:** High-efficiency photovoltaic module designed for optical

- wireless power receiver charging via targeted laser beams.
 - Wireless inductive charging (Qi standard).
- **"mini retake" System (Reserve & Generation):**
 - **Thermoelectric Generator (TEG):** Seebeck element installed on the inner housing surface leveraging the high thermal conductivity of the titanium base. Converts ambient-to-body thermal gradients into background trickling charge.
 - **Hardware Reserve (10%):** An isolated, dedicated power circuit inaccessible to regular processing tasks.
 - **"mini retake RESET" Button:** Mechanical override switch to force-release reserve power, re-energizing the NPU and LoRa transmitter during critical main-battery exhaustion.

5. COMMUNICATIONS, CRYPTOGRAPHY, AND EMERGENCY BEACONING

- **High-Speed LiFi Link:**
 - Transceiver capable of up to 10 Gbps throughput via an optical port for localized data exchange with responder tablets, system log offloading, and AI model synchronization.
- **LoRa / Mesh RF Module (Sub-GHz) & Antenna Array:**
 - Semtech SX1262 transceiver operating at 868/915 MHz. Effective range: 3–5 km in dense forest, up to 25 km line-of-sight.
 - **Titanium Shielding Mitigation:** A radio-transparent window made of zirconia ceramics / PEEK is integrated into the top stamped segment. The retractable 316L stainless steel chassis functions as an external railing antenna.
 - **Beacon Mode:** Automated broadcast of binary telemetry packets (ID, coordinates, heart rate, SpO_2) every 5 minutes across the Mesh network.
- **BLE 5.2/5.3 with Hardware Encryption:**
 - User profile provisioning (full name, blood group, allergies) via mobile application.
 - End-to-end hardware encryption utilizing AES-128 (CryptoCell) and ECDH P-256 key exchange (MITM attack protection).
 - The BLE radio is hardware-gated to cut power during "Wilderness Mode" operation.

6. ENCLOSURE MANUFACTURING, ERGONOMICS, AND INGRESS PROTECTION

- **Housing Fabrication (Hot Isothermal Stamping):**
 - Primary load-bearing segments are produced via hot isothermal stamping of Grade 5 titanium alloy at 850–900°C under an argon shield gas atmosphere.
 - Precision CNC post-machining for O-ring channels, display seats, and sensor apertures.
- **Ingress Protection (IP68 / IP69K):**
 - Sealing achieved by laser welding structural covers post-assembly of internal PCBs, complemented by liquid silicone rubber (LSR) fluorosilicone gaskets at display and optical interfaces.
 - Acoustic and gas sampling microchannels are protected with press-fit ePTFE

membranes.

- **Dimensions and Mass:** Monolithic 40 mm wide cuff form factor. Top segment thickness: 15 mm; bottom segment thickness: 10–12 mm. Total calculated assembly weight: $\sim 160\text{--}180\text{ g}$.
- **User Interface (UI/UX):**
 - High-contrast display embedded in the top segment, protected by mineral glass with an impact-absorbing rubber bumper frame.
 - Front-facing and lateral status LED indicators following a standard green/yellow/red diagnostic convention.
 - High-intensity SOS strobe light mounted on the outer lateral edge.
 - UI card navigation driven by voice commands ("Next"/"Back") or IMU motion/flick gestures.

7. EXTENDED SURVIVAL DECISION MATRIX

No.	Scenario Name (Disaster Class)	Input Triggers (Sensors / Acoustics / Voice)	AI Output Instructions (Display / Haptics / Audio)
1	Severe Bleeding / Trauma	Voice: "I'm bleeding", "Deep cut" Sensors: $\text{HR} > 120\text{ bpm}$, GSR stress spike	Display: "1. Apply pressure above the wound. 2. Apply pressure bandage. 3. Elevate the limb."
2	Hypothermia	Voice: "Freezing" Sensors: $\text{Air Temp} < -10^{\circ}\text{C}$, $\text{Skin Temp} < 34.5^{\circ}\text{C}$	Display: "1. Seek shelter from the wind. 2. Insulate your body from the ground. 3. Rub your chest and move your fingers."
3	Heat Stroke / Dehydration	Voice: "Heat exhaustion" Sensors: $\text{Air Temp} > +38^{\circ}\text{C}$, $\text{Skin Temp} > 37.8^{\circ}\text{C}$	Display: "1. Move into the shade. 2. Dampen cloth and apply to forehead. 3. Sip water slowly."
4	Disorientation / Lost	Voice: "I'm lost" Sensors: Erratic IMU movement profile	Display: Full-screen digital compass display. "Head North. Look for streams and clearings."
5	Emergency Fall	Sensors: $\text{IMU impact event} > 4\text{ g}$ + $\text{immobility} > 60\text{ seconds}$	Haptics: 30-second countdown timer. If unacknowledged: trigger visual/audio SOS beacon + transmit LoRa Beacon packet.
6	Avalanche / Seismic Threat	Acoustics: Infrasound rumble (1–15 Hz)	Haptics: Heavy continuous vibration

No.	Scenario Name (Disaster Class)	Input Triggers (Sensors / Acoustics / Voice)	AI Output Instructions (Display / Haptics / Audio)
		Sensors: Micro-seismic IMU tremors, rapid barometric drop	pattern. Display: "1. Evacuate dangerous slope/riverbed. 2. Cover airways. 3. Seek shelter behind rock formations."
7	High-Pressure Gas Leak	Acoustics: High-frequency ultrasonic hiss ($>20\text{ kHz}$) Sensors: TVOC/ CO_2 concentration spike	Haptics: Rapid short pulses. Display: "1. Avoid open flames! 2. Move crosswind. 3. Put on protective mask."
8	UAV / Vehicle Approach	Acoustics: Harmonic motor frequency profile Sensors: Micro-vibrations via IMU	Haptics: Double short pulse. Display: "1. Clear open areas. 2. Seek cover under dense canopy/structure."
9	Wild Predators (Night Watch)	Acoustics: Heavy footsteps, snapping branches, growls Sensors: GSR peak (resting panic)	Haptics: Gentle alert pulse. Display: "1. Do not run! 2. Make yourself look larger (raise arms). 3. Back away slowly."

8. TORUS PROJECT PRODUCT LINE AND ECOSYSTEM ROADMAP

-  **TORUS Urban:** Lightweight stamped aluminum/titanium housing designed for everyday use, focused on non-invasive health tracking and high-speed LiFi synchronization.
-  **TORUS Extreme (Wild Nature):** IP69K rated heavy-duty titanium monocoque casing equipped with infrasound avalanche and wildlife monitoring alongside LoRa Mesh support.
-  **TORUS Military / Rescue:** High-impact titanium construction featuring a non-reflective matte DLC coating, UAV detection algorithms, RF anomaly sensing, and low-observable tactical display modes.
-  **TORUS Medical / Care:** Designed for automated stroke and seizure detection with emergency dispatch integration for emergency services and caregivers.
-  **TORUS Junior:** Wearable tailored for children featuring geo-fencing capabilities and environmental hazard monitoring.